Weak solution of the merged mathematical equations of the polluted atmosphere

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Abstract

Considered as a geophysical fluid, the polluted atmosphere shares the shallow domain characteristics with other natural large-scale fluids such as seas and oceans. This means that its domain is excessively greater horizontally than in the vertical dimension, leading to the classic hydrostatic approximation of the Navier-Stokes equations. The authors of the [Azérad and Guillén, 2001] article have proved an existence theorem for this model with respect to the ocean, without considering pollution effects. In this present work we provide a generalization of their result, extending the fluid velocity equations with an additional convection-diffusion equation representing pollutants in the atmosphere.

Keywords: Navier-Stokes equations, shallow domains, pollution evolution equation, hydrostatic approximation, compactness, weak solutions.

Contents

1 Introduction

We investigate the existence of a weak solution for the merged motion-pollution atmosphere model. At this point the functions we work with are $\mathbf{v} = (\mathbf{v}_H, v_3)$, and C, representing velocity and concentration, respectively (we are not considering other variables such as water vapor quantity or temperature).

We underline the fact that this work is for the case of **inland** scenario. Even though the [Azérad and Guillén, 2001] article works with the equations of the ocean, this work is the extension of that article in the sense that we transfer and exploit its main result for the case of the atmosphere, and for this reason it is the main reference article for this work.

2 The equations describing the polluted atmosphere

The first step is to choose a coordinate system, set the domain we start with and an appropriate set of equations describing both atmosphere motions and the effect of pollution. The set of equations, the choice of the locally acceptable Cartesian coordinate system we are using for the velocity terms are similar to those suggested by our reference article. We ignore large scale effects such as the curvature of the Earth. For describing the effect of pollution we use a continuity equation written in Cartesian coordinates for the concentration.

We fix the local Cartesian coordinate system to be adjusted to the downwind direction, ie. z is the vertical direction, while x is the downwind, y is the crosswind direction. It is a legitimate thing to do if we are considering a time interval which is not too long in the sense that it is reasonable to assume that we have a permanent, relatively stable main wind direction. It is true that this is a certain restriction on the model, but as we will see later, exploiting this choice will help us simplify the model a bit.

Let us consider the following domain to begin with. This domain is a local slice of the atmosphere filled with homogeneous air that we assume to be incompressible, and a pollutant of concentration C.

$$\Omega_\epsilon = \{(x, y, z) \in \mathbb{R}^3; (x, y) \in \omega, 0 < z < \epsilon h(x, y)\}, \tag{1}$$

where ω is a Lipschitz-domain in $\mathbb{R}^2$, and h is a nonnegative Lipschitzian application.

The equations describing the air velocity in the atmosphere are the incompressible Navier-Stokes equations:

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v} - \Delta_\nu \mathbf{v} + \nabla q + 2\mathbf{w} \times \mathbf{v} = \mathbf{g} \text{ in } \Omega_\epsilon \times (0, T) \tag{2}$$

$$\nabla \cdot \mathbf{v} = 0 \text{ in } \Omega_\epsilon \times (0, T) \tag{3}$$

Contaminants emitted into the atmosphere are transported by the wind, mixed and dispersed by turbulent behavior, and deposited onto the ground by gravity. The dispersion of pollutants is commonly modeled by a continuity equation, and this is what we will use to describe the pollution phenomena as well.

$$\partial_t C + \mathbf{v} \cdot \nabla C = K_x \frac{\partial^2 C}{\partial x^2} + K_y \frac{\partial^2 C}{\partial y^2} + K_z \frac{\partial^2 C}{\partial z^2} + S \tag{4}$$

In this equation S represents a source term. From now on we will assume that S is as regular as needed.

3 Rescaling

3.1 Rescaling due to shallow domain

$$x = x_1, y = x_2, z = \epsilon x_3$$
$$v_1 = u_1^\epsilon, v_2 = u_2^\epsilon, v_3 = \epsilon u_3^\epsilon, p = p^\epsilon$$
$$\nu_x = \nu_1, \nu_y = \nu_2, \nu_z = \epsilon^2 \nu_3$$
$$K_z = \epsilon^2 K_3$$

So we have a new domain

$$\Omega = \{(x_1, x_2, x_3) \in \mathbb{R}^3; (x_1, x_2) \in \omega, 0 < x_3 < h(x_1, x_2)\},$$

3.2 Rescaling due to a special choice of coordinate system

Here we exploit the fact that we chose the coordinate system in a way so that the x axis matches the downwind direction. In this case we use an additional scaling equation, which is different in nature from the previous ones in the sense that it is not derived from the shallowness of the domain but rather suggested by the fact that in the downwind direction it is natural to assume that the diffusion is negligible compared to downwind transport, ie.

$$|u_1 \partial_{x_1} C| \gg \left| K_x \frac{\partial^2 C}{\partial x_1^2} \right| \tag{5}$$

which leads us to using

$$K_x = \epsilon K_1$$

Putting this condition and the previously described rescaling terms together to the original equations, we arrive to the merged anisotropic equations of the polluted atmosphere above inland surface.

3.3 Rescaled merged equations - Anisotropic Navier-Stokes plus pollution

$$\partial_t u_1^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_1^\epsilon - \Delta_\nu u_1^\epsilon - \alpha u_2^\epsilon + \epsilon\beta u_3^\epsilon + \partial_1 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{6}$$

$$\partial_t u_2^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_2^\epsilon - \Delta_\nu u_2^\epsilon + \alpha u_1^\epsilon + \partial_2 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{7}$$

$$\epsilon^2 \{\partial_t u_3^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon - \Delta_\nu u_3^\epsilon\} - \epsilon\beta u_1^\epsilon + \partial_3 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{8}$$

$$\nabla \cdot \mathbf{u}^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{9}$$

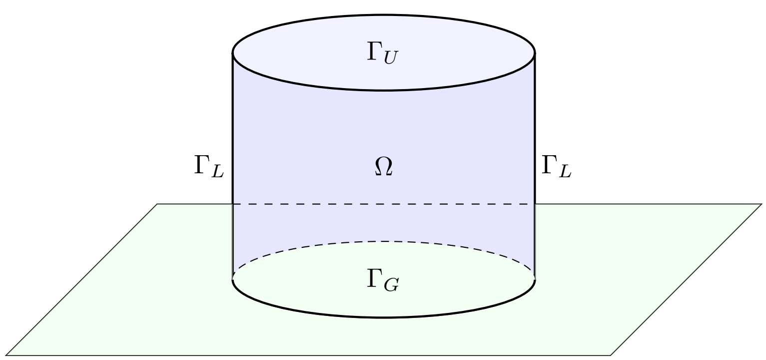
$$\mathbf{u}^\epsilon = 0 \text{ on } \Gamma \times (0,T) \tag{10}$$

$$\mathbf{u}^\epsilon(\cdot, t=0) = \mathbf{u}_0, C^\epsilon(\cdot, t=0) = C_0^\epsilon \text{ in } \Omega \tag{11}$$

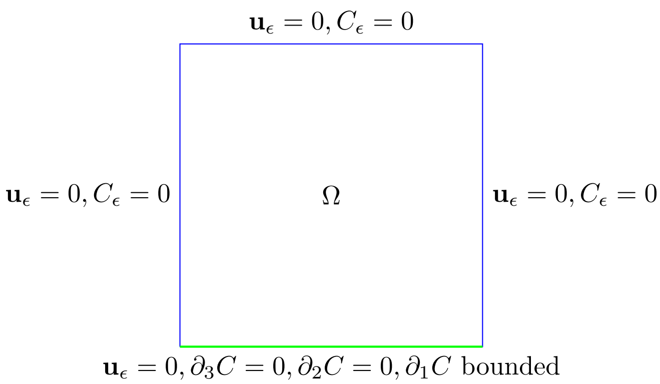
$$\partial_t C^\epsilon + \mathbf{u}^\epsilon \cdot \nabla C^\epsilon = \epsilon K_1 \frac{\partial^2 C^\epsilon}{\partial x_1^2} + K_2 \frac{\partial^2 C^\epsilon}{\partial x_2^2} + K_3 \frac{\partial^2 C^\epsilon}{\partial x_3^2} + S \tag{12}$$

4 Boundary conditions

It is vital to choose boundary conditions for the pollution concentration that are physically as accurate as possible and at the same time not unnecessarily complicated.



The boundary division of the $\partial\Omega$: the upper boundary of the Ω domain is denoted by Γ_U, the lateral boundary is Γ_L, and the lower boundary of the domain is Γ_G. As we mentioned before, the case we are investigating is that of inland domains Ω, ie the case of landscapes and cities with no significant nearby water surface. This is very important since the boundary conditions for both wind velocity and pollution concentration in the air are different depending whether we are above solid ground or water surface. The boundary conditions



we choose to use in the inland scenario are the following, shown in a cross-section.

An explanation of the boundary conditions:

- The upper boundary of the domain we are considering is chosen to be at the isobar representing the planetary boundary layer ([Arya, 2001]), on this $p = p_0$ pressure level the upper boundary of the local domain is denoted by Γ_U. On Γ_U we use $\mathbf{u}_H = 0, u_3 = 0, C(x, y, z) = 0$ boundary conditions.

- On both sides the lateral boundary of the rectangle shaped Ω domain is given by Γ_L. Here we use the same boundary conditions as on the upper level, ie. we have $\mathbf{u}_H = 0, u_3 = 0, C(x, y, z) = 0$.

  Remark. *If we wanted to use $C(x, y, z) = c$ with some c constant value, we could make a change of variables distracting that background flow and arrive back to the same mathematical problem with C being zero.*

- In this model we neglect topological variations, and the $z = 0$ plane represents the lower boundary. On the ground level we have $u_H = 0, u_3 = 0$ boundary conditions for the air velocity. This is the mathematical way of expressing no-slip boundary conditions for the horizontal wind speed and the impervious nature of the ground with respect to the wind. As for the concentration, the way we can choose ground level boundary conditions is the following. Firstly, we assume that the pollution is not absorbed by the ground, so we use $\partial_3 C = 0$ on Γ_G (see also for example [Yeh and Huang, 1975]). For the horizontal partial derivatives of the pollution, we can separate downwind and crosswind direction boundary conditions. In the crosswind direction we use $\partial_2 C = 0$ while in the downwind (stronger) direction our assumption is more relaxed, we only assume that $\partial_1 C$ is finite in the boundary norm, ie in $L^2 L^2(0, T; \Gamma_G)$.

Remark. *In the paper of [Temam and Ziane, 2005], on p35 it is suggested that the domain we work on has the form of $p_0 < p < p_1$, because for the* *thin portion of air between the earth and the isobar $p = p_1$, another specific simplified model would be necessary. So a way to make the model more precise is to cut the domain at an isobar a little bit off the physical ground. This would of course radically change the boundary conditions we use for the functions, because it would take away the advantage of having a wall-like, natural and physical boundary condition for our domain to support it. If we are positively above the ground, $\partial_3 C = 0$ would represent a strange reflective layer.*

5 Hydrostatic Navier-Stokes plus pollution

$$\partial_t u_1 + \mathbf{u} \cdot \nabla u_1 - \Delta_\nu u_1 - \alpha u_2 + \partial_1 p = 0 \text{ in } \Omega \times (0, T) \tag{13}$$

$$\partial_t u_2 + \mathbf{u} \cdot \nabla u_2 - \Delta_\nu u_2 - \alpha u_1 + \partial_2 p = 0 \text{ in } \Omega \times (0, T) \tag{14}$$

$$\partial_3 p = 0 \text{ in } \Omega \times (0, T) \tag{15}$$

$$\nabla \cdot \mathbf{u} = 0 \text{ in } \Omega \times (0, T) \tag{16}$$

$$u_1 = u_2 = u_3 n_3 = 0 \text{ on } \Gamma \times (0, T) \tag{17}$$

$$u_i(\cdot, t = 0) = u_{0i}, C(\cdot, t = 0) = C_0 \text{ in } \Omega, i = 1,2 \tag{18}$$

$$\partial_t C + \mathbf{u} \cdot \nabla C = K_2 \frac{\partial^2 C}{\partial x_2^2} + K_3 \frac{\partial^2 C}{\partial x_3^2} + S \tag{19}$$

6 Main theorem

- The article of [Azérad and Guillén, 2001] proved an existence theorem for weak solution of the equations describing the ocean without pollution. We can use a slightly modified version of this, translated to the inland atmosphere domain.

  Theorem 1. *Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0, \mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; there exists a weak solution $\mathbf{u}$ of the hydrostatic Navier-Stokes equations as the aspect ratio ϵ tends to 0.*

- The task we want to do is to extend this result in the sense that it becomes valid to the merged set of $(\mathbf{u}, C)$ equations as well.

  Theorem 2. *Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0, \mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; and let $C(0) \in L^2(\Omega)^3$, $C(0) = 0$ on $\partial\Omega$, there exists a weak solution ($\mathbf{u}$, C) of the hydrostatic Navier-Stokes equations as the aspect ratio ϵ tends to 0.*

7 Weak formulation

We need to calculate the weak formulation of the merged equations in both the anisotropic and hydrostatic case. Since this inland atmosphere model has slightly different boundary conditions even in the velocity equations than the model of the ocean, we will see that the weak formulation of our system has minor differences from the ocean scenario.

7.1 Weak formulation of the hydrostatic system

For the weak formulation we will use the modified version of the spaces defined in the original article.

$$H^1_\Gamma(\Omega) = \{v \in H^1(\Omega); v = 0 \text{ on } \partial\Omega = \Gamma\}$$

$$\mathbf{V} = \{\mathbf{v} \in H^1_\Gamma(\Omega) \times H^1_\Gamma(\Omega) \times H^1_0(\Omega); \nabla \cdot \mathbf{v} = 0 \text{ in } \Omega\}$$

$$H(\partial_3, \Omega) = \{v \in L^2(\Omega); \partial_3 v \in L^2(\Omega)\}$$

$$H_0(\partial_3, \Omega) = \{v \in H(\partial_3, \Omega); vn_3 = 0 \text{ on } \partial\Omega\}$$

$$\mathbf{W} = \{\mathbf{u} \in H^1_\Gamma(\Omega) \times H^1_\Gamma(\Omega) \times H_0(\partial_3, \Omega); \nabla \cdot \mathbf{u} = 0 \text{ in } \Omega\}$$

Find $(\mathbf{u}, C)$ such that $\mathbf{u} = (\mathbf{u}_H, u_3) \in L^2(0, T; \mathbf{W})$ with $\mathbf{u}_H \in L^\infty(0, T; L^2(\Omega)^2)$, and C with $\partial_2 C, \partial_3 C \in L^2_t L^2_x$, $C \in L^\infty L^2$ and $C(0) \in L^2_x$, such that

$$\begin{aligned}
&\int_0^T -(\mathbf{u}_H, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H, (\mathbf{u} \cdot \nabla)\mathbf{v}_H) + (b(\mathbf{u}_H), \mathbf{v}_H) dt \\
&+ \int_0^T -(C, \partial_t \tilde{C}) - (\mathbf{u}C, \nabla\tilde{C}) + K_2(\partial_2 C, \partial_2 \tilde{C}) + K_3(\partial_3 C, \partial_3 \tilde{C}) dt = \\
&= -(\mathbf{u}_{0H}, \mathbf{v}_H(0)) - (C(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C}) dt
\end{aligned} \tag{20}$$

for all $(\mathbf{v}, \tilde{C})$ with $\mathbf{v} = (\mathbf{v}_H, v_3) \in H^1(0, T, \mathbf{W})$, with $v_H(T) = 0$ and $\partial_3 \mathbf{v}_H \in L^\infty(0, T; L^3(\Omega)^2)$ and $\tilde{C}$ with $\tilde{C} \in L^2(0, T, H^1)$, $\tilde{C} \in H^1(0, T, L^2)$, $\tilde{C}(T) = 0$.

7.2 Weak formulation of the anisotropic system

Find $\mathbf{u}^\epsilon = (\mathbf{u}_H, u_3^\epsilon) \in L^2(0, t; \mathbf{V}) \cap L^\infty(0, T; L^2(\Omega)^3)$ and C with $C \in L^\infty L^2$, $C \in L^2(0, T; H^1)$, $C(0) \in L^2_x$ such that

$$
\begin{aligned}
&\int_0^T -(\mathbf{u}_H^\epsilon, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H^\epsilon, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H^\epsilon, (\mathbf{u}^\epsilon \cdot \nabla)\mathbf{v}_H) + (b(\mathbf{u}_H^\epsilon), \mathbf{v}_H) dt \\
&+ \epsilon\beta\big[\int_0^T (u_3^\epsilon, v_1) - (u_1^\epsilon, v_3)\big] dt \\
&+ \epsilon^2\big[\int_0^T -(u_3^\epsilon, \partial_t v_3) + (\mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon, v_3) + (\nabla_\nu u_3^\epsilon, \nabla_\nu v_3) dt\big] \\
&+ \int_0^T -(C^\epsilon, \partial_t \tilde{C}^\epsilon) - (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}^\epsilon) + \\
&\epsilon K_1(\partial_1 C^\epsilon, \partial_1 \tilde{C}^\epsilon) - \epsilon K_1\big[(\partial_1 C^\epsilon)\tilde{C}\big]_{\Gamma_G} + K_2(\partial_2 C^\epsilon, \partial_2 \tilde{C}^\epsilon) + K_3(\partial_3 C^\epsilon, \partial_3 \tilde{C}^\epsilon) dt = \\
&= -(\mathbf{u}_{0H}^\epsilon, \mathbf{v}_H^\epsilon(0)) - \epsilon^2(u_{03}^\epsilon, v_3^\epsilon(0)) - (C^\epsilon(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C}) dt
\end{aligned}
\tag{21}
$$

for all $\mathbf{v} = (\mathbf{v}_H, v_3) \in H^1(0, T; \mathbf{V})$ with $\mathbf{v}(T) = 0$ and $\tilde{C}$ such that $\tilde{C} \in H^1(0, T, H^1)$ and $\tilde{C}(T) = 0$.

To pass to the limit in the terms that do not include pollution in the anisotropic equation has already been worked out in the [Azérad and Guillén, 2001] article. What we need to work on is how to legitimately pass to the limit with the C terms. To do this our first key tool will be the energy inequality. Once we have a priori estimates for the functions in the linear terms, we can pass to the limit with those without particular problems; the challenging part will be to prove convergence for the $(\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}^\epsilon)$ term, here we will need a stronger result than just the weak convergence of the present functions in the product.

8 Energy inequality and a priori estimates

8.1 Energy inequality

Calculating the energy inequality for the merged system we arrive to

$$
\begin{aligned}
&\frac{1}{2}(\|\mathbf{u}_H^\epsilon(t)\|_{L^2}^2 + \epsilon^2\|u_3^\epsilon(t)\|_{L^2}^2 + \|C^\epsilon(t)\|_{L^2}^2) + \int_0^t \{\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2\|\nabla_\nu u_3^\epsilon\|_{L^2}^2\} d\tau + \\
&\int_0^t \{\epsilon K_1\|\partial_1 C^\epsilon\|_{L^2}^2 + K_2\|\partial_2 C^\epsilon\|_{L^2}^2 + K_3\|\partial_3 C^\epsilon\|_{L^2}^2\} d\tau \leq \\
&\frac{1}{2}(\|\mathbf{u}_H^\epsilon(0)\|_{L^2}^2 + \epsilon^2\|u_3^\epsilon(0)\|_{L^2}^2 + \|C^\epsilon(0)\|_{L^2}^2) + \int_0^t (S, C^\epsilon) d\tau + \int_0^t \epsilon K_1\langle \partial_1 C^\epsilon, C^\epsilon\rangle_{\Gamma_G} d\tau
\end{aligned}
\tag{22}
$$

In order to use the energy inequality to extract a priori estimates, we need to know that the right hand side of the inequality is bounded. Since the S source term is assumed to be as smooth as we need it to be, the only term we need to work further is $\int_0^t \epsilon K_1 \langle \partial_1 C^\epsilon, C^\epsilon\rangle_{\Gamma_G} d\tau$.

We can estimate this term in the following way.

$$
\begin{aligned}
&\int_0^t \epsilon K_1 \langle \partial_1 C^\epsilon, C^\epsilon\rangle_{\Gamma_G} d\tau \le K_1 \int_0^t \sqrt{\epsilon}\|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)} \sqrt{\epsilon}\|C^\epsilon\|_{L^2(\Gamma_G)} \le \\
&\frac{1}{\delta} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C^\epsilon\|^2_{L^2(\Gamma_G)} + \delta K_1 \int_0^t \frac{\epsilon}{2}\|C^\epsilon\|^2_{L^2(\Gamma_G)} \le \\
&B.T. + \delta \frac{\epsilon}{2} K_1 c_T \int_0^t \|C^\epsilon\|^2_{W^{1,2}(\Omega)} ,
\end{aligned} \tag{23}
$$

where c_T is the constant provided by the trace theorem, and $B.T.$ denotes the boundary term which is bounded because of the a priori assumptions made on boundary conditions. Now we can carry on with the estimate by using the $W^{1,2}$ norm definition, exploit the fact that we are on a bounded domain and then applying the Sobolev inequality (for $n = 3, p = 2, p* = 6$, denoting its constant by c_S), so we have

$$
\begin{aligned}
&\int_0^t \epsilon K_1 \langle \partial_1 C^\epsilon, C^\epsilon\rangle_{\Gamma_G} d\tau \le \\
&B.T. + \delta \frac{\epsilon}{2} K_1 c_T \int_0^t \{\|C^\epsilon\|^2_{L^2(\Omega)} + \|\nabla C^\epsilon\|^2_{L^2(\Omega)}\} d\tau \le \\
&B.T. + \delta \frac{\epsilon}{2} K_1 c_T \int_0^t \{\|C^\epsilon\|^2_{L^6(\Omega)} + \|\nabla C^\epsilon\|^2_{L^2(\Omega)}\} d\tau \le \\
&B.T. + \delta \frac{\epsilon}{2} K_1 c_T \int_0^t \{c_S\|\nabla C^\epsilon\|^2_{L^2(\Omega)} + \|\nabla C^\epsilon\|^2_{L^2(\Omega)}\} d\tau = \\
&B.T. + \delta \frac{\epsilon}{2} K_1 c_T (c_S+1) \int_0^t \|\nabla C^\epsilon\|^2_{L^2(\Omega)} d\tau
\end{aligned} \tag{24}
$$

Which means that the energy inequality takes the form of

$$
\begin{aligned}
&\frac{1}{2}(\left\|\mathbf{u}^\epsilon_H(t)\right\|^2_{L^2} + \epsilon^2\left\|u^\epsilon_3(t)\right\|^2_{L^2} + \left\|C^\epsilon(t)\right\|^2_{L^2}) + \int_0^t \{\|\nabla_\nu \mathbf{u}^\epsilon_H\|^2_{L^2} + \epsilon^2\|\nabla_\nu u^\epsilon_3\|^2_{L^2}\} d\tau + \\
&\int_0^t \{(\epsilon K_1 - \delta\frac{\epsilon}{2} K_1 c_T (c_S+1))\|\partial_1 C^\epsilon\|^2_{L^2} + (K_2 - \delta\frac{\epsilon}{2} K_1 c_T(c_S+1))\|\partial_2 C^\epsilon\|^2_{L^2} + \\
&+ (K_3 - \delta\frac{\epsilon}{2} K_1 c_T (c_S+1))\|\partial_3 C^\epsilon\|^2_{L^2}\} d\tau \le \\
&\frac{1}{2}(\left\|\mathbf{u}^\epsilon_H(0)\right\|^2_{L^2} + \epsilon^2\left\|u^\epsilon_3(0)\right\|^2_{L^2} + \left\|C^\epsilon(0)\right\|^2_{L^2}) + \int_0^t (S, C^\epsilon) d\tau + \frac{1}{\delta} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C^\epsilon\|^2_{L^2(\Gamma_G)}
\end{aligned} \tag{25}
$$

The right hand side of the energy inequality is now apparently bounded, and it is easy to see that the coefficients of the pollution terms are positive, since the δ constant of the generalized Young inequality can be freely chosen in a way such that the requirements $\epsilon K_1(1-\delta\frac{1}{2}c_T(c_S+1)) > 0, (K_2-\delta\frac{\epsilon}{2}K_1 c_T(c_S+1)) > 0, (K_3-\delta\frac{\epsilon}{2}K_1 c_T(c_S+1)) > 0$ are all satisfied.

8.2 A priori estimates and weak convergence

Now we can extract a priori estimates from the final form of the energy inequality. We have

- $L_t^\infty L_x^2$ boundedness for $u_1, u_2, \epsilon u_3$,
- $L_t^2 H_x^1$ boundedness for $u_1, u_2, \epsilon u_3$,
- $L_t^\infty L_x^2$ boundedness for C,
- $L_t^2 L_x^2$ boundedness for $\sqrt{\epsilon}\partial_1 C, \partial_2 C, \partial_3 C$.

Because of the boundedness we have weak convergence, ie. there are subsequences still denoted by the same way, for which we have $C^\epsilon \rightharpoonup C$ in $L_t^\infty L_x^2$, $\partial_2 C^\epsilon \rightharpoonup \partial_2 C$ and $\partial_3 C^\epsilon \rightharpoonup \partial_3 C$ in $L_t^2 L_x^2$, and $\sqrt{\epsilon}\partial_1 C^\epsilon \rightharpoonup e$ for some limit element e in $L_t^2 L_x^2$.

9 Passing to the limit

Passing to the limit is done without problems for the terms that do not include pollution term, since that has been proved before in the [Azérad and Guillén, 2001] article. In the linear terms the weak convergence guaranteed by the a priori estimates will be enough. In the nonlinear term we need an additional result that provides a weak convergence of the product of functions for which we only had weak convergence separately.

9.1 Linear terms

For the linear terms we can directly apply the weak convergence results of the previous section.

(i) The convergence of the $(C^\epsilon, \frac{\partial \tilde{C}}{\partial t})$ term is easy to see, since we have $C^\epsilon \rightharpoonup C$ in $L_t^\infty L_x^2$.

(ii) For the term we have $\epsilon K_1 \int_0^T (\partial_1 C^\epsilon, \partial_1 \tilde{C}) = \sqrt{\epsilon} K_1 \int_0^T (\sqrt{\epsilon}\partial_1 C^\epsilon, \partial_1 \tilde{C})$, which goes to 0 as $\epsilon \to 0$ because of the Cauchy-Schwartz inequality, since we have $\sqrt{\epsilon}\partial_1 C^\epsilon$ bounded in $L_t^2 L_x^2$, and $\tilde{C}$ is a test function.

(iii) As we have seen before, we have $\partial_2 C^\epsilon \rightharpoonup \partial_2 C$ in $L_t^2 L_x^2$, which by definition provides that

$$\int_0^T (\partial_2 C^\epsilon, f)\mathrm{d}t \to \int_0^T (\partial_2 C, f)\mathrm{d}t \qquad \forall f \in L_t^2 L_x^2.$$

(iv) Analogously,

$$\int_0^T (\partial_3 C^\epsilon, f)\mathrm{dt} \to \int_0^T (\partial_3 C, f)\mathrm{dt} \qquad \forall f \in L^2_t L^2_x.$$

(v) The $\epsilon K_1 \int_0^T \left[(\partial_1 C^\epsilon)\tilde{C}\right]_{\Gamma_G} \mathrm{dt}$ line integral vanishes as $\epsilon \to 0$ since by the Cauchy-Schwartz equation we can estimate the integral by two bounded terms ($\partial_1 C^\epsilon$ was assumed to be finite in norm on the lower boundary).

9.2 Nonlinear term

We need some sort of compactness theorem to handle $(\mathbf{u}^\epsilon C^\epsilon, \nabla\tilde{C})$.

Remark. *As a matter of fact, we actually have three terms here, namely $(u_i^\epsilon C^\epsilon, \partial_i \tilde{C})$ for $i = 1, 2, 3$, and for the values $1, 2$ of i the weak convergence of the $u_i^\epsilon C^\epsilon$ product is given by the fact that, as it is proved in the reference article, we actually have strong convergence for the horizontal velocities, and because of the a priori estimate from the energy bound we know we have weak convergence for the C^ϵ term. But this argument does not really solve the challenge caused by nonlinearity because it does not hold for u_3, so we still need an additional compactness theorem as mentioned before.*

In this case the Aubin-Lions lemma can not be applied since it would require for C^ϵ to be in a compactly embedded space such as L^2H^1, which is not the case here because we only know boundedness results for $\sqrt{\epsilon}\partial_1 C$, but not for $\partial_1 C$.

We will show that the following lemma from [Lions, 1996] is applicable in our case.

Lemma 1. *Let g_n, h_n converge weakly to g, h respectively in $L^{p_1}(0,T; L^{p_2}(\Omega))$, $L^{q_1}(0,T; L^{q_2}(\Omega))$ where $1 \leq p_1, p_2 \leq +\infty$,*

$$\frac{1}{p_1} + \frac{1}{q_1} = \frac{1}{p_2} + \frac{1}{q_2} = 1.$$

We assume in addition that

- *$\frac{\partial g^n}{\partial t}$ is bounded in $L^1(0,T; W^{-m,1}(\Omega))$ for some $m \geq 0$ independent of n.*
- *$\left\|h^n - h^n(\cdot + \xi, t)\right\|_{L^{q_1}(0,T;L^{q_2}(\Omega))} \to 0$ as $|\xi| \to 0$, uniformly in n.*

Then, $g^n h^n$ converges to gh (in the sense of distributions on $\Omega \times (0,T)$).

What we will show in the following is that for the functions u_i^ϵ and C^ϵ the assumptions of Lemma 1 are satisfied, which means that $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$ holds.

Lemma 2. *We have $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$.*

Proof. We have seen from the a priori estimates that $C^\epsilon \rightharpoonup C$ in $L^\infty_t L^2_x$, and we know from the reference article that $u_i^\epsilon \rightharpoonup u_i$ in $L^2_t L^2_x$.

Let's choose $q_1 = q_2 = p_1 = p_2 = 2$. This is fine since we are on a finite domain, so $C^\epsilon \in L^\infty L^2$ implies $C^\epsilon \in L^2 L^2$.

- **Bound for the time derivative**.

$$\frac{\partial C^\epsilon}{\partial t} = -u^\epsilon \nabla C^\epsilon + \sqrt{\epsilon}\, \underbrace{K_1\partial_1 \underbrace{\partial_1 \sqrt{\epsilon} C^\epsilon}_{L^2}}_{W^{-1,2}} + \underbrace{K_2\partial_2 \underbrace{\partial_2 C^\epsilon}_{L^2}}_{W^{-1,2}} + \underbrace{K_3\partial_3 \underbrace{\partial_3 C^\epsilon}_{L^2}}_{W^{-1,2}} - S$$

Since the S source term was assumed to be very regular, we see that the second part of the right hand side is in $W^{-1,2}$. On the other hand, the first part is in $W^{-1,1}$, because we have

$$u^\epsilon \nabla C^\epsilon = \underbrace{\nabla \cdot (\underbrace{u^\epsilon C^\epsilon}_{L^1})}_{W^{-1,1}} - \underbrace{\nabla \cdot (u^\epsilon)}_{0} C.$$

Eventually we have $\frac{\partial C}{\partial t} \in L^1 W^{-1,1}$.

- **Equicontinuity**. We need to show that $\left\| u_i^n - u_i^n(\cdot + \xi, t) \right\|_{L^2 L^2} \to 0$ as $\xi \to 0$. Since in the time domain the condition naturally holds, we will focus on the space part of the proof.

First we show that for a function f in the function space C_0^∞ the condition holds.

$$\left\| f(x+\xi) - f(x) \right\|_{L^2} = (\int_\Omega \left| f(x+\xi) - f(x) \right|^2 dx)^{1/2} \leq$$

$$\left(\int_\Omega (\sup_x \left| f(x+\xi) - f(x) \right|)^2 d\mathrm{x} \right)^{1/2} = \left\| f(x+\xi) - f(x) \right\|_{L^\infty} \cdot |\Omega|^{1/p}.$$

So we have

$$\lim_{\xi \to 0} \left\| f(x+\xi) - f(x) \right\|_{L^2} \leq |\Omega|^{1/p} \cdot \lim_{\xi \to 0} (\sup_x \left| f(x+\xi) - f(x) \right|) = 0,$$

since $f \in C_0^\infty$.

As a second step we show that the condition holds for any function $f \in L^2$. Here we exploit the fact that C_0^∞ is dense in the L^p function space. By definition this means that $\exists \varphi_n \in C_0^\infty(\Omega)$ such that $\varphi_n \to f$ in $L^p(\Omega)$. With this we have

$$\left\| f(x+\xi) - f(x) \right\|_{L^2} =$$

$$\left\| f(x+\xi) - \varphi_n(x+\xi) + \varphi_n(x+\xi) - \varphi_n(x) + \varphi_n(x) - f(x) \right\|_{L^2} \leq$$

$$\left\| f(x+\xi) - \varphi_n(x+\xi) \right\|_{L^2} + \left\| \varphi_n(x+\xi) - \varphi_n(x) \right\|_{L^2} + \left\| \varphi_n(x) - f(x) \right\|_{L^2}$$

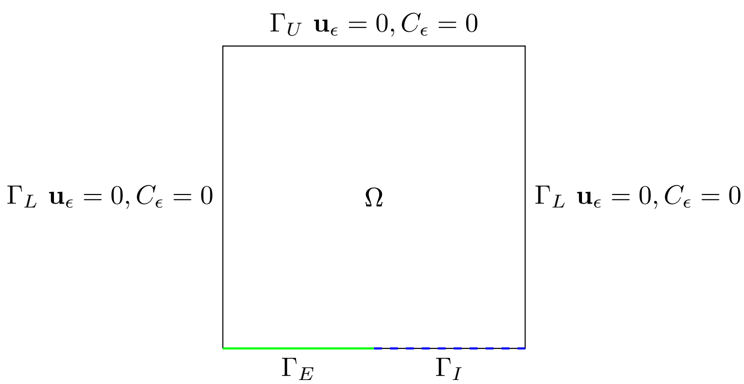
The first and third term goes to 0 as $\xi \to 0$, since $\varphi_n \to f$ in $L^p(\Omega)$. On the other hand in the first part of the proof we have shown exactly that the second term goes to 0. Which proves that $\left\|u_i^n - u_i^n(\cdot+\xi,t)\right\|_{L^2L^2} \to 0$ as $\xi \to 0$. Eventually this means that we can apply the lemma with $h^n = u_i^\epsilon, g^n = C^\epsilon$, which proves the statement of the lemma.

$\square$

10 A generalized model

We now will present a generalization of the previous model by applying the following modifications.

- Drop the assumption about the coordinate system being fixed onto the main wind direction, ie. drop the ϵ from K_x, —it comes from a specifically chosen coordinate system.
- Adapt the model for coastal conditions.



On the ground layer we naturally separate the boundary into two different parts in this model, $\Gamma_G = \Gamma_E \bigcup \Gamma_I$; Γ_E denoting the part of the boundary above the ground, and Γ_I denoting the section that is interacting with the sea. The separation is a sort of expansion or generalization of the ground boundary since while above ground parts it is reasonable to use no-slip conditions, above the ocean it is natural to assume that the wind has an interaction with the waves, there is no such rigidness as in the ground case. For this case we will include θ, which represents the wind traction on the ocean surface. For the pollution concentration as well, above sea territory it is not natural to use $\partial_3 C = 0$, since the particles can now sink under the boundary level.

On Γ_E we have the same boundary conditions as in the inland case, $\mathbf{u}^\epsilon = 0, \partial_3 C = 0, \partial_2 C = 0, \partial_1 C$ bounded. On Γ_I on the other hand we have $\nu_3 \partial_3 \mathbf{u}_H^\epsilon = \theta_H, u_3 = 0$ and $\partial_1 C, \partial_2 C, \partial_3 C$ bounded in the boundary norm, similarly to what we saw in the inland model.

We will discuss in more details the parts that are different from the inland case.

10.1 Weak formulation

The weak formulation of the coastal generalized systems will be different from the original inland version in that *a)* it will contain a boundary term for the wind because of the air - ocean interface, *b)* it will contain boundary terms for $\partial_2 C, \partial_3 C$, which had been assumed to be zero before on the ground, and finally *c)* here the $\partial_1 C$ terms are present that were absent in the previous version because of the ϵ used in K_x. The terms which are new or different in this version are underlined.

10.1.1 Anisotropic system

$$\begin{aligned}
&\int_0^T -(\mathbf{u}_H^\epsilon, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H^\epsilon, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H^\epsilon, (\mathbf{u}^\epsilon \cdot \nabla)\mathbf{v}_H) + (b(\mathbf{u}_H^\epsilon), \mathbf{v}_H) dt \\
&+ \epsilon\beta\big[\int_0^T (u_3^\epsilon, v_1) - (u_1^\epsilon, v_3)\big] dt \\
&+ \epsilon^2\big[\int_0^T -(u_3^\epsilon, \partial_t v_3) + (\mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon, v_3) + (\nabla_\nu u_3^\epsilon, \nabla_\nu v_3) dt\big] \\
&+ \int_0^T -(C^\epsilon, \partial_t \tilde{C}^\epsilon) - (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}^\epsilon) + \underline{K_1}(\partial_1 C^\epsilon, \partial_1 \tilde{C}^\epsilon) - \underline{K_1}\big[(\partial_1 C^\epsilon)\tilde{C}\big]_{\Gamma_G} + \\
&K_2(\partial_2 C^\epsilon, \partial_2 \tilde{C}^\epsilon) - \underline{K_2\big[(\partial_2 C^\epsilon)\tilde{C}\big]_{\Gamma_I}} + K_3(\partial_3 C^\epsilon, \partial_3 \tilde{C}^\epsilon) - \underline{K_3\big[(\partial_3 C^\epsilon)\tilde{C}\big]_{\Gamma_I}} dt = \\
&= -(\mathbf{u}_{0H}^\epsilon, \mathbf{v}_H^\epsilon(0)) - \underline{\int_0^T \langle \theta_H, \mathbf{v}_H \rangle_{\Gamma_I}} - \epsilon^2 (u_{03}^\epsilon, v_3^\epsilon(0)) - (C^\epsilon(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C}) dt
\end{aligned} \tag{26}$$

10.1.2 Hydrostatic system

$$\begin{aligned}
&\int_0^T -(\mathbf{u}_H, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H, (\mathbf{u} \cdot \nabla)\mathbf{v}_H) + (b(\mathbf{u}_H), \mathbf{v}_H) dt \\
&+ \int_0^T -(C, \partial_t \tilde{C}) - (\mathbf{u}C, \nabla \tilde{C}) + \underline{K_1(\partial_1 C, \partial_1 \tilde{C}) - K_1\big[(\partial_1 C)\tilde{C}\big]_{\Gamma_G}} + \\
&K_2(\partial_2 C, \partial_2 \tilde{C}) - \underline{K_2\big[(\partial_2 C)\tilde{C}\big]_{\Gamma_I}} + K_3(\partial_3 C, \partial_3 \tilde{C}) - \underline{K_3\big[(\partial_3 C)\tilde{C}\big]_{\Gamma_I}} dt = \\
&= -(\mathbf{u}_{0H}, \mathbf{v}_H(0)) - \underline{\int_0^T \langle \theta_H, \mathbf{v}_H \rangle_{\Gamma_I}} - (C(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C}) dt
\end{aligned} \tag{27}$$

10.2 Energy inequality

The analogous terms that we saw in the weak formulation will be different in the energy inequality as well. The coastal, generalized version of the energy inequality will be the following. The different parts are again noted by underline.

$$
\begin{aligned}
&\frac{1}{2}(\left\|\mathbf{u}_H^\epsilon(t)\right\|_{L^2}^2 + \epsilon^2\left\|u_3^\epsilon(t)\right\|_{L^2}^2 + \left\|C^\epsilon(t)\right\|_{L^2}^2) + \int_0^t \{\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2\|\nabla_\nu u_3^\epsilon\|_{L^2}^2\} d\tau + \\
&\int_0^t \{\underline{K_1}\|\partial_1 C^\epsilon\|_{L^2}^2 + K_2\|\partial_2 C^\epsilon\|_{L^2}^2 + K_3\|\partial_3 C^\epsilon\|_{L^2}^2\} d\tau \leq \\
&\frac{1}{2}(\left\|\mathbf{u}_H^\epsilon(0)\right\|_{L^2}^2 + \epsilon^2\left\|u_3^\epsilon(0)\right\|_{L^2}^2 + \left\|C^\epsilon(0)\right\|_{L^2}^2) + \int_0^t (S, C^\epsilon) d\tau - \underline{\int_0^t \langle \theta_H, \mathbf{u}_H\rangle_{\Gamma_I}} \\
&\int_0^t \underline{K_1}\langle \partial_1 C^\epsilon, C^\epsilon\rangle_{\Gamma_G} + \underline{K_2\langle \partial_2 C^\epsilon, C^\epsilon\rangle_{\Gamma_I}} + \underline{K_3\langle \partial_3 C^\epsilon, C^\epsilon\rangle_{\Gamma_I}} d\tau
\end{aligned} \tag{28}
$$

Analogously as we estimated the boundary term in the energy inequality before, we can do the estimate

$$
\int_0^t K_i \langle \partial_i C^\epsilon, C^\epsilon\rangle_{\Gamma_I} d\tau \leq \frac{1}{\delta}\frac{K_i}{2}\int_0^t \|\partial_i C^\epsilon\|_{L^2(\Gamma_I)}^2 + \delta\frac{K_i}{2} c_T(c_S+1)\int_0^t \|\nabla C^\epsilon\|_{L^2(\Omega)}^2 d\tau
$$

for $i = 1, 2, 3$. The first term for every value of i is ab ovo bounded, so it causes no trouble on the right hand side. The term containing the wind is also assumed to be bounded.

Eventually we have

$$
\begin{aligned}
&\frac{1}{2}(\left\|\mathbf{u}_H^\epsilon(t)\right\|_{L^2}^2 + \epsilon^2\left\|u_3^\epsilon(t)\right\|_{L^2}^2 + \left\|C^\epsilon(t)\right\|_{L^2}^2) + \int_0^t \{\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2\|\nabla_\nu u_3^\epsilon\|_{L^2}^2\} d\tau + \\
&\int_0^t \underline{\{(K_1 - \delta\frac{K_1+K_2+K_3}{2} c_T(c_S+1))\|\partial_1 C^\epsilon\|_{L^2}^2 +} \\
&\underline{(K_2 - \delta\frac{K_1+K_2+K_3}{2} c_T(c_S+1))\|\partial_2 C^\epsilon\|_{L^2}^2 +} \\
&\underline{(K_3 - \delta\frac{K_1+K_2+K_3}{2} c_T(c_S+1))\|\partial_3 C^\epsilon\|_{L^2}^2\} d\tau} \leq \\
&\frac{1}{2}(\left\|\mathbf{u}_H^\epsilon(0)\right\|_{L^2}^2 + \epsilon^2\left\|u_3^\epsilon(0)\right\|_{L^2}^2 + \left\|C^\epsilon(0)\right\|_{L^2}^2) + \int_0^t (S, C^\epsilon) d\tau - \underline{\int_0^t \langle \theta_H, \mathbf{u}_H\rangle_{\Gamma_I}} \\
&\int_0^t \underline{\frac{1}{\delta_1}\frac{K_1}{2}\int_0^t \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)}^2} + \underline{\frac{1}{\delta_2}\frac{K_2}{2}\int_0^t \|\partial_2 C^\epsilon\|_{L^2(\Gamma_I)}^2} + \underline{\frac{1}{\delta_3}\frac{K_3}{2}\int_0^t \|\partial_3 C^\epsilon\|_{L^2(\Gamma_I)}^2} d\tau
\end{aligned} \tag{29}
$$

Even though by first look the inequality seems different from the inland one, it basically provides the same estimates and weak convergences. We have

- $L_t^\infty L_x^2$ boundedness for $u_1^\epsilon, u_2^\epsilon, \epsilon u_3^\epsilon$,

- $L_t^2 H_x^1$ boundedness for $u_1^\epsilon, u_2^\epsilon, \epsilon u_3^\epsilon$,
- $L_t^\infty L_x^2$ boundedness for C^ϵ,
- $L_t^2 L_x^2$ boundedness for $\partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$, basically we have $L_t^2 H_x^1$ boundedness for C^ϵ.

10.3 Passing to the limit

As for passing to the limit, everything is very similar to the inland version, except for the terms $K_i\big[(\partial_i C^\epsilon)\tilde{C}\big]_{\Gamma_I}$.

$$\int_0^T K_i\big[(\partial_i C^\epsilon - \partial_i C)\tilde{C}\big]_{\Gamma_I} dt \leq$$
$$\delta K_i \int_0^T \|\partial_i C^\epsilon - \partial_i C\|^2_{L^2(\Gamma_I)} + \frac{1}{\delta} K_i c_T (c_S + 1) \int_0^T \big\|\tilde{C}\big\|^2_{L^2(\Gamma_I)}$$

Since $\tilde{C}$ is a test function, we can assume that the second term disappears. If we assume $\partial_i C^\epsilon$ is a given constant on the lower boundary, then it would work.

11 Future work, remarks

- We mentioned a small ambiguity regarding where to cut the lower boundary of the domain we are considering. Other authors use the $z = 0$ simplified ground level cut as well, such as [Hosseini, 2013].
- Use more sophisticated ground-level boundary conditions instead of $\partial_3 C = 0$ in the inland case, for example [Monin, 1959].
- We could consider multiple contaminants.
- It is possible and not unreasonable in certain cases to use isotropic diffusivities $K_x = K_y = K_z$.
- It is option to simply take $K_x = 0$, without using ϵ as suggested by [Hosseini, 2013], advection / convection effect being dominant compared to diffusion. It could also be possible to assume that $K_x = K_z$ due to symmetry (p16).

12 TODO

1.) Include B Hosseini thesis in the reference list, include 4 physics book in the reference list.

2.) $\mathbf{u}C$ product to be in L^2 in the hydrostatic weak formulation - spaces.

3.) trace theorem can be used with $\|u\|_{W^{1,p}(U)}$ if u is in that space, but in the inland case, C^ϵ is not in H^1 in space.

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